



# Federated Architecture

In this lesson, we will discuss federated architecture.

We'll cover the following



- What Is federated architecture?
- How is federated architecture implemented in decentralized social networks?
- What is the need for pods?

## What Is federated architecture?

Federated architecture is an extension of decentralized architecture. It powers social networks like Mastodon, Minds, Diaspora, etc.

The term federated in a general sense means a group of semi-autonomous entities that exchange information with each other. A real-world example of this is looking at different states in a country managed by the state



things running smoothly. Then, those state governments share information with each other and with a central government making a complete autonomous government.

This is just an example. From a technical standpoint, the federated model is under continual research, development, and evolution. There are no standard rules. Developers and architects can have their own designs in place.

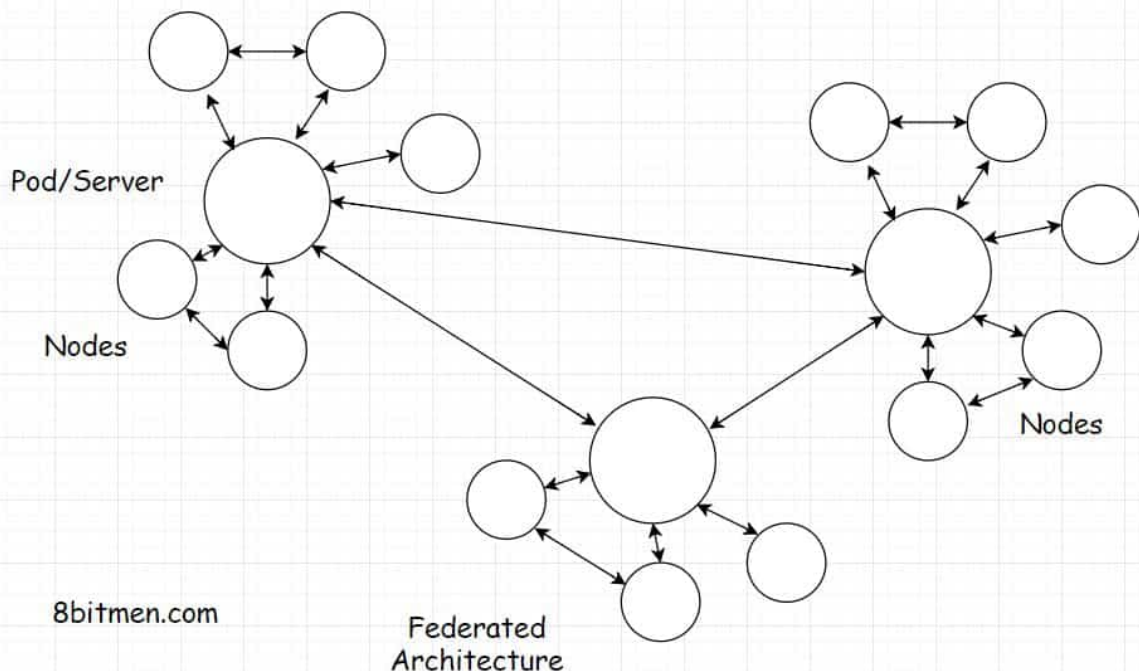


# How is federated architecture implemented in decentralized social networks?

> As shown in the illustration below, a federated network has entities called servers or pods. A large number of nodes subscribe to the pods. There are several pods in the network that are linked to each other and share information.

The pods can be hosted by individuals and as new pods are hosted and introduced to the network, the network keeps growing.

In case the link between a few pods breaks temporarily, the network is still up. Nodes can still communicate with each other via the pods they are subscribed to.



What is the need for Pods? Can't the nodes just be linked to each other like in a regular peer-to-peer network?

# What is the need for pods?

Pods facilitate node discovery. In a peer-to-peer network, there is no way of discovering other nodes, and we would just sit in the dark if it weren't for a centralized node registry or something.

The other way is to run a scan through the network to discover other nodes. This is a time-consuming and tedious task. Why not just have a pod instead?

Well, I believe you now have a fundamental understanding of the p2p architecture and the decentralized web. Let's move on to the next lesson, where we talk about picking the right server-side technology.

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